



2012 Assessment Examination

FORM VI

MATHEMATICS EXTENSION 2

Thursday 17th May 2012

General Instructions

- Writing time — 1 Hour 30 minutes
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 55 Marks

- All questions may be attempted.

Section I – 7 Marks

- Questions 1–7 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II – 48 Marks

- Questions 8–11 are of equal value.
- All necessary working should be shown in every question.
- Start each question in a new booklet.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Candidature — 86 boys

Collection

- Write your candidate number clearly on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Eight.
- Write your candidate number on this question paper and submit it with your answers.

Examiner

BDD

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

Which of the following expressions is a primitive of $\frac{1}{\sqrt{4-9x^2}}$?

1

(A) $\frac{1}{3} \sin^{-1} \frac{3x}{2}$

(B) $\frac{1}{9} \sin^{-1} \frac{3x}{2}$

(C) $\sin^{-1} \frac{3x}{2}$

(D) $\frac{1}{2} \sin^{-1} \frac{3x}{2}$

QUESTION TWO

The remainder when $16x^4 - 8x + 3$ is divided by $x + i$ is:

1

(A) $-13 + 8i$

(B) $19 + 8i$

(C) $-13 - 8i$

(D) $19 - 8i$

QUESTION THREE

The angle between the asymptotes of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ is approximately:

1

(A) 74°

(B) 37°

(C) 61°

(D) 29°

Exam continues next page ...

QUESTION FOUR

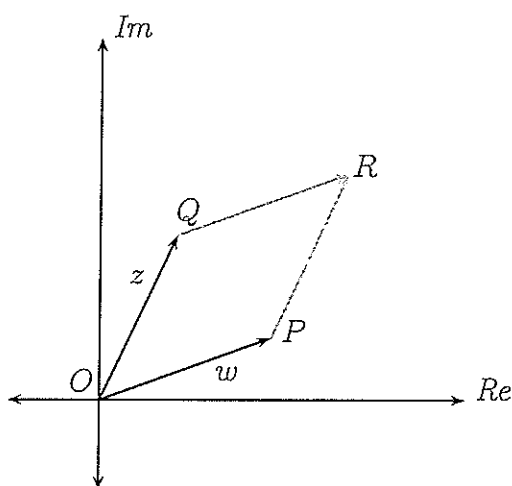
At the point $(2 \cos \theta, \sin \theta)$ on the ellipse $x^2 + 4y^2 = 4$ the tangent has gradient:

1

- (A) $\frac{1}{2} \cot \theta$
- (B) $\frac{1}{2} \tan \theta$
- (C) $-\frac{1}{2} \cot \theta$
- (D) $-2 \tan \theta$

QUESTION FIVE

1



Let P and Q represent complex numbers w and z respectively in the first quadrant of the Argand Plane, with $\arg z > \arg w$, as in the diagram above. The diagram also shows the addition parallelogram. Then $w - z$ is represented by the vector:

- (A) $\overrightarrow{PQ}$
- (B) $\overrightarrow{RP}$
- (C) $\overrightarrow{OR}$
- (D) $\overrightarrow{QP}$

QUESTION SIX

One root of the quadratic equation $iz^2 + 3z + 3 - 11i = 0$ is $3 + 2i$. The second root is:

1

- (A) $-2i$
- (B) $-3 + i$
- (C) $-6 - 2i$
- (D) $-3 - 5i$

QUESTION SEVEN

A skydiver falls out of a plane which is flying horizontally. Initially his motion is determined by the acceleration due to gravity 10 m/s^2 and any resistance is negligible. After 5 seconds, he opens his parachute and his motion is determined by the equation $\ddot{x} = 10 - \frac{v}{4}$. His terminal velocity is 40 m/s . Downwards is taken as positive.

1

Which statement best reflects the situation after he opens his parachute?

- (A) He hits the ground with a vertical speed of 40 m/s .
- (B) His vertical speed never exceeds 40 m/s .
- (C) His vertical speed never drops below 40 m/s .
- (D) We need to know his horizontal speed to complete this question.

————— End of Section I —————

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION EIGHT (12 marks) Use a separate writing booklet.

Marks

- (a) Using a suitable substitution, or otherwise, evaluate $\int_0^{\frac{\pi}{3}} \sec^2 x \tan x \, dx$. 2
- (b) By a suitable use of a Pythagorean identity, find $\int \sin^3 x \cos^2 x \, dx$. 2
- (c) Consider the ellipse $\mathcal{E}$ with equation $9x^2 + 16y^2 = 144$.
- (i) Find its eccentricity. 1
 - (ii) Write down the coordinates of the foci. 1
 - (iii) Write down the equations of the directrices. 1
 - (iv) Sketch the ellipse, showing any intercepts with the axes and the information found above. 2
- (d) It is known that the equation $x^3 + 3x^2 + 7x + 5 = 0$ has roots α , β and γ .
- (i) Find a simplified monic polynomial equation with roots $\alpha + 1$, $\beta + 1$ and $\gamma + 1$. 2
 - (ii) Hence solve $x^3 + 3x^2 + 7x + 5 = 0$. 1

QUESTION NINE (12 marks) Use a separate writing booklet.

Marks

- (a) Use a t -substitution to evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \cos x + 2 \sin x} \, dx$ 4
- (b) Let $P(x) = 2x^3 + (1 - 4k)x^2 + (2k^2 - 2k)x + k^2$.
- (i) Show that $x = k$ is a double root of $P(x) = 0$. 2
 - (ii) Hence find the third root of $P(x) = 0$. 1
- (c) Consider the hyperbola $\mathcal{H}$ with equation $xy = c^2$ where c is a positive constant and let $P(cp, \frac{c}{p})$ be a variable point on $\mathcal{H}$. Suppose that the tangent at P intersects the x -axis at A and let M be the midpoint of AP .
- (i) Show that the tangent to $\mathcal{H}$ at P has equation $x + p^2y = 2cp$. 2
 - (ii) Show that M has coordinates $(\frac{3}{2}cp, \frac{c}{2p})$. 2
 - (iii) Find the Cartesian equation of the locus of M , as P moves along the hyperbola $\mathcal{H}$. 1

Exam continues overleaf ...

QUESTION TEN (12 marks) Use a separate writing booklet.

Marks

(a) Let $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$.

(i) Prove that $I_n = \frac{n-1}{n} I_{n-2}$, for $n \geq 2$.

3

(ii) Find I_4 .

1

(b) An object with unit mass is launched straight upwards with an initial speed of u m/s.It experiences the force of gravity and also a retarding force of $\frac{1}{40}v^2$ Newtons. Let x be its vertical displacement from its point of launch and take upwards as positive and the acceleration due to gravity as $g = 10\text{m/s}^2$.

(i) Show that the equation of motion is $\ddot{x} = \frac{-400 - v^2}{40}$.

1

(ii) Show that the object rises to a maximum height of $20 \ln \left(\frac{400 + u^2}{400} \right)$.

2

(iii) With an initial speed of 10 m/s the particle reaches a maximum height of $20 \ln \frac{5}{4}$ metres. What launch speed would double this maximum height? Working is required.

2

(c) Suppose that $0 < x < \frac{1}{4}$.

(i) Prove that $1 - 2x > 2x > 0$.

1

(ii) Find $\sin \cos^{-1} \frac{2x}{2x-1}$.

2

QUESTION ELEVEN (12 marks) Use a separate writing booklet.

Marks

(a) Let $I = \int_0^{\frac{\pi}{2}} \ln \sin x \, dx$.

(i) Use the substitution $u = \frac{\pi}{2} - x$ to show that

1

$$I = \int_0^{\frac{\pi}{2}} \ln \cos x \, dx.$$

(ii) Use the substitution $u = \pi - x$ to show that

2

$$I = \frac{1}{2} \times \int_0^{\pi} \ln \sin x \, dx.$$

(iii) Use part (i) to show that $2I = \int_0^{\frac{\pi}{2}} \ln \left(\frac{1}{2} \sin 2x \right) dx$.

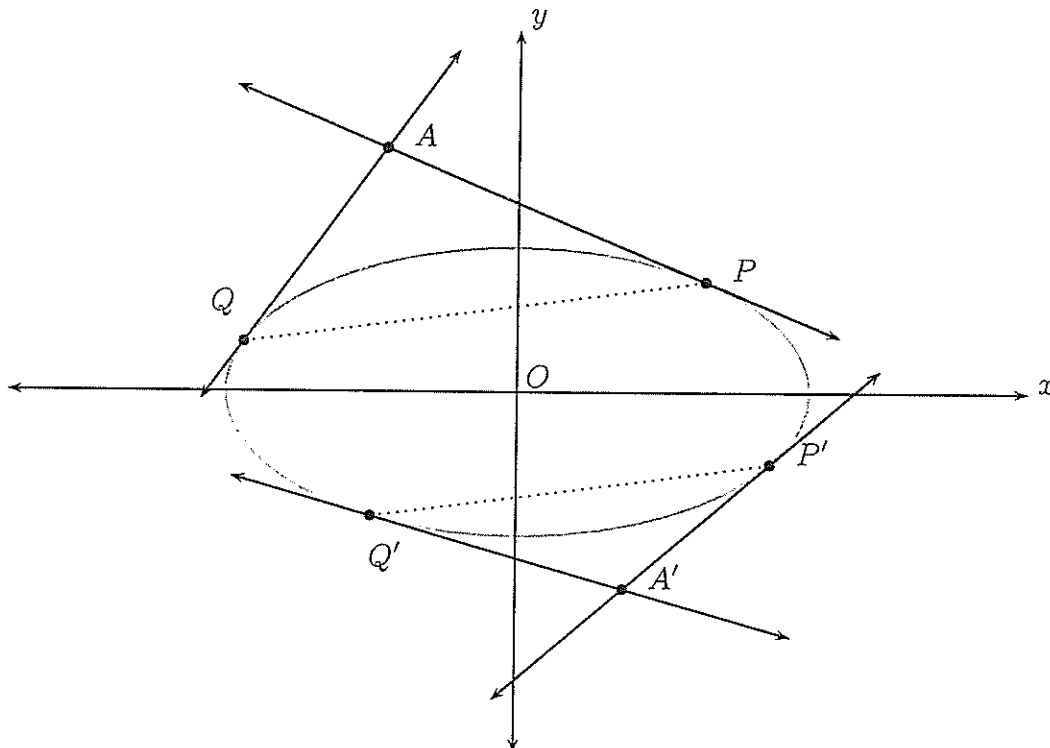
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(iv) Hence use parts (ii) and (iii) to show that $I = -\frac{\pi}{2} \ln 2$.

2

Exam continues next page ...

(b)



Consider the ellipse $\mathcal{E}$ with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $b < a$. Let $O(0,0)$ be the centre of the ellipse.

Let PQ be the chord of contact defined by the point $A(x_0, y_0)$ external to the ellipse such that $P(x_1, y_1)$ and $Q(x_2, y_2)$ lie on $\mathcal{E}$.

Similarly let $P'Q'$ be the chord of contact defined by the point $A'(x'_0, y'_0)$ external to the ellipse such that $P'(x'_1, y'_1)$ and $Q'(x'_2, y'_2)$ lie on $\mathcal{E}$.

- (i) Show that the tangent to $\mathcal{E}$ at $P(x_1, y_1)$ has equation $\frac{x_1x}{a^2} + \frac{y_1y}{b^2} = 1$. 2
- (ii) Assume the the chord of contact PQ has equation $\frac{x_0x}{a^2} + \frac{y_0y}{b^2} = 1$. Show that if 2
 A , A' and O are collinear, then the chords PQ and $P'Q'$ are parallel.
- (iii) If the chords PQ and $P'Q'$ are not parallel, then suppose they intersect at 2
 $R(x_3, y_3)$. Prove that the chord of contact from R lies on the line AA' .

————— End of Section II —————

END OF EXAMINATION

SECTION I - Multiple Choice**QUESTION ONE**

The required indefinite integral is:

$$\begin{aligned}\int \frac{1}{\sqrt{4-9x^2}} &= \frac{1}{2} \int \frac{1}{\sqrt{1-\left(\frac{3x}{2}\right)^2}} \\ &= \frac{1}{2} \times \frac{2}{3} \sin^{-1}\left(\frac{3x}{2}\right)\end{aligned}$$

Hence A.

QUESTION TWO

Let $P(x) = 16x^4 - 8x + 3$. Then $P(-i) = 19 + 8i$. Hence B.

QUESTION THREE

The angle of inclination of the asymptote $y = \frac{b}{a}x$ is $\tan^{-1}\left(\frac{3}{4}\right) \doteq 37^\circ$. The angle between the asymptotes is twice this, hence A.

QUESTION FOUR

By parametric differentiation, $\frac{dy}{dx} = \frac{\cos \theta}{-2 \sin \theta}$, hence C.

QUESTION FIVE

The vector $w - z$ runs from Q to P , hence D.

QUESTION SIX

Let the second root be α . Then the sum of the roots is $\alpha + 3 + 2i = 3i$, thus $\alpha = -3 + i$. Hence B.

QUESTION SEVEN

After 5 seconds, the skydiver is travelling at 50 m/s, i.e. faster than the terminal velocity. Once he opens his parachute, his speed will slow to approach the terminal velocity, but never drop below this limiting velocity. Hence C.

SECTION II - Written Response

QUESTION EIGHT (12 marks)

Marks

(a)

2

$$\begin{aligned}\int_0^{\frac{\pi}{3}} \sec^2 x \tan x \, dx &= \left[\frac{1}{2} \tan^2 x \right]_0^{\frac{\pi}{3}} \\ &= \frac{1}{2} (\tan^2 \frac{\pi}{3} - \tan^2 0) \\ &= \frac{3}{2}\end{aligned}$$

(b)

2

$$\begin{aligned}\int \sin^3 x \cos^2 x \, dx &= \int \sin x (1 - \cos^2 x) \cos^2 x \, dx \\ &= \left(\int \sin x \cos^2 x - \sin x \cos^4 x \right) dx \\ &= \frac{-\cos^3 x}{3} + \frac{\cos^5 x}{5} + C\end{aligned}$$

(c) (i) We have $b^2 = a^2(1 - e^2)$ so $e^2 = 1 - \frac{b^2}{a^2}$.

1

Since $a = 4$, $b = 3$ we have $e^2 = 1 - \frac{9}{16} = \frac{7}{16}$.Hence $e = \frac{\sqrt{7}}{4}$.(ii) The foci are $(\pm ae, 0) = (\pm\sqrt{7}, 0)$.

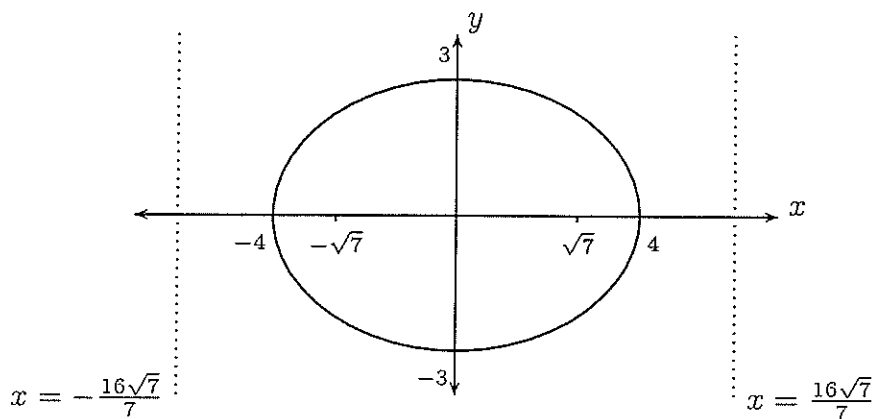
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(iii) The directrices are $x = \frac{a}{e}$ or $x = -\frac{a}{e}$. That is, $x = \frac{16\sqrt{7}}{7}$ or $x = -\frac{16\sqrt{7}}{7}$.

1

(iv)

2



- (d) (i) Let $P(x) = x^3 + 3x^2 + 7x + 5 = 0$. The required polynomial is $P(x - 1)$. The equation is: 2

$$\begin{aligned}(x - 1)^3 + 3(x - 1)^2 + 7(x - 1) + 5 &= 0 \\ x^3 - 3x^2 + 3x - 1 + 3(x^2 - 2x + 1) + 7x - 7 + 5 &= 0 \\ x^3 + 4x &= 0 \\ x(x^2 + 4) &= 0\end{aligned}$$

- (ii) This new equation has roots $x = 0, x = \pm 2i$. Hence the original polynomial equation has roots $-1, -1 \pm 2i$. 1

QUESTION NINE (12 marks)

Marks

(a)

4

Let $t = \tan \frac{x}{2}$

Then $x = 2 \tan^{-1} t$

Thus $dx = \frac{2dt}{1+t^2}$

$$\begin{aligned}\text{Hence } \int_0^{\frac{\pi}{2}} \frac{1}{1 + \cos x + 2 \sin x} dx &= \int_0^1 \frac{1}{1 + \frac{1-t^2}{1+t^2} + \frac{4t}{1+t^2}} \frac{2}{1+t^2} dt \\ &= \int_0^1 \frac{2}{1+t^2 + 1-t^2 + 4t} dt \\ &= \int_0^1 \frac{2}{2+4t} dt \\ &= \int_0^1 \frac{1}{1+2t} dt \\ &= \left[\frac{1}{2} \ln(1+2t) \right]_0^1 \\ &= \frac{1}{2} \ln 3\end{aligned}$$

(b) (i)

2

$$\begin{aligned}\text{We have } P(k) &= 2k^3 + (1-4k)k^2 + (2k^2-2k)k + k^2 \\ &= 2k^3 + k^2 - 4k^3 + 2k^3 - 2k^2 + k^2 \\ &= 0\end{aligned}$$

$$\begin{aligned}\text{and } P'(k) &= 6k^2 + (1-4k)2k + (2k^2-2k) \\ &= 6k^2 + 2k - 8k^2 + 2k^2 - 2k \\ &= 0\end{aligned}$$

Hence k is a double root of $P(x)$.

- (ii) Since the product of the roots is $-\frac{k^2}{2}$, the third root must be $-\frac{1}{2}$. 1

- (c) (i) By parametric differentiation,

2

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy/dp}{dx/dp} \\ &= \frac{-\frac{c}{p^2}}{-c} \\ &= -\frac{1}{p^2}\end{aligned}$$

Hence at the point $P(cp, \frac{c}{p})$ the tangent is

$$\begin{aligned}(y - \frac{c}{p}) &= -\frac{1}{p^2}(x - cp) \\ p^2y - cp &= -x + cp \\ x + p^2y &= 2cp\end{aligned}$$

- (ii) The
- x
- intercept has coordinates
- $A(2cp, 0)$
- . Point
- M
- is the midpoint of
- AP
- , hence it has coordinates

2

$$\left(\frac{1}{2}(2cp + p), \frac{1}{2}(0 + \frac{c}{p})\right) = \left(\frac{3}{2}cp, \frac{c}{2p}\right)$$

- (iii) Eliminating the parameter
- p
- is easily done by noting

1

$$\begin{aligned}xy &= \frac{3}{2}cp \times \frac{c}{2p} \\ xy &= \frac{3}{4}c^2\end{aligned}$$

This is a hyperbola— simply a stretched version of the original.

QUESTION TEN (12 marks)**Marks**

- (a) (i) Using integration by parts,

3

$$\begin{aligned}I_n &= \int_0^{\frac{\pi}{2}} \cos^n x \, dx \\ &= \int_0^{\frac{\pi}{2}} \cos x \times \cos^{n-1} x \, dx \\ &= \left[\sin x \cos^{n-1} x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x \times -\sin x (n-1) \cos^{n-2} x \, dx \\ &= 0 + (n-1) \times \int_0^{\frac{\pi}{2}} \sin^2 x \cos^{n-2} x \, dx \\ &= (n-1) \times \int_0^{\frac{\pi}{2}} (1 - \cos^2 x) \cos^{n-2} x \, dx \\ &= (n-1) \times \left(\int_0^{\frac{\pi}{2}} \cos^{n-2} x \, dx - \int_0^{\frac{\pi}{2}} \cos^n x \, dx \right) \\ &= (n-1) \times (I_{n-2} - I_n)\end{aligned}$$

$$\text{Hence } I_n + (n-1)I_n = (n-1)I_{n-2}$$

$$nI_n = (n-2)I_{n-1}$$

$$I_n = \frac{n-1}{n} I_{n-2}$$

(ii)

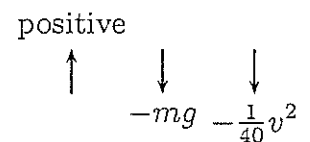
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$$\begin{aligned} I_4 &= \frac{3}{4} I_2 \\ &= \frac{3}{4} \times \frac{1}{2} \times \int_0^{\frac{\pi}{2}} dx \\ &= \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} \\ &= \frac{3\pi}{16} \end{aligned}$$

(b) (i)

1

The forces on the object are gravity $-mg$ and a retarding force opposing the motion $-\frac{1}{40}v^2$. By Newton's second law, the resulting force gives equation $m\ddot{x} = -mg - \frac{1}{40}v^2$. Taking $g = 10$ and $m = 1$ (since the object has unit mass) gives



$$\begin{aligned} \ddot{x} &= -10 - \frac{1}{40}v^2 \\ &= \frac{-400 - v^2}{40} \end{aligned}$$

(ii) Solving this equation of motion gives:

2

$$\begin{aligned} \ddot{x} &= \frac{-400 - v^2}{40} \\ v \frac{dv}{dx} &= -\frac{400 + v^2}{40} \\ -\frac{40v dv}{400 + v^2} &= dx \end{aligned}$$

$$\begin{aligned} \text{Hence } \int dx &= -20 \times \int \frac{2v}{400 + v^2} dv \\ x &= -20 \ln(400 + v^2) + C \end{aligned}$$

Since $x = 0$ when $v = u$, we have $C = 20 \ln(400 + u^2)$. The maximum height $x = H$ is attained when $v = 0$, so

$$\begin{aligned} H &= -20 \ln 400 + 20 \ln(400 + u^2) \\ &= 20 \ln \left(\frac{400 + u^2}{400} \right) \end{aligned}$$

- (iii) We need to find what initial speed u gives a maximum height $40 \ln \frac{5}{4}$. That is, we must solve 2

$$40 \ln \frac{5}{4} = 2 \ln \left(\frac{400 + u^2}{400} \right)$$

$$2 \ln \frac{5}{4} = \ln \left(\frac{400 + u^2}{400} \right)$$

$$\ln \frac{25}{16} = \ln \left(\frac{400 + u^2}{400} \right)$$

$$\frac{25}{16} = \frac{400 + u^2}{400}$$

$$625 = 400 + u^2$$

$$225 = u^2$$

$$u = 15$$

- (c) (i)

1

$$0 < x < \frac{1}{4}$$

$$0 < 2x < \frac{1}{2}$$

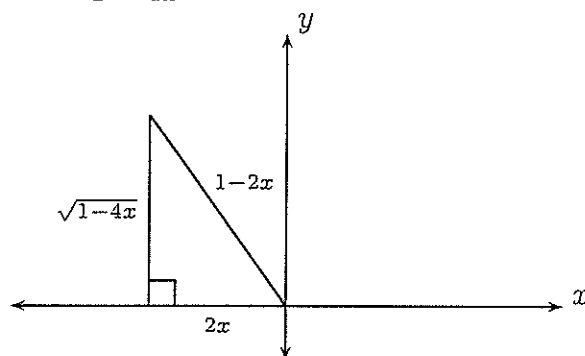
$$0 > -2x > -\frac{1}{2}$$

$$1 > 1 - 2x > \frac{1}{2}$$

But $\frac{1}{2} > 2x > 0$, so $1 - 2x > 2x > 0$.

- (ii) Let $\theta = \cos^{-1} \frac{2x}{2x-1}$. Since $\frac{2x}{2x-1} < 0$, the angle θ is in quadrant 2. The related triangle is drawn below: note that $|2x-1| = 1-2x > 0$. The missing side length ℓ in this triangle is given by 2

$$\begin{aligned} \ell^2 &= (1-2x)^2 - (2x)^2 \\ &= 1-4x \end{aligned}$$



The angle is in quadrant two, hence $\sin \theta = +\frac{\sqrt{1-4x}}{1-2x}$.

QUESTION ELEVEN (12 marks)**Marks**

(a) (i)

1

$$\begin{aligned}
 I &= \int_0^{\frac{\pi}{2}} \ln \sin x \, dx \\
 &= \int_{\frac{\pi}{2}}^0 \ln \sin\left(\frac{\pi}{2} - u\right) \times -du \\
 &= \int_0^{\frac{\pi}{2}} \ln \sin\left(\frac{\pi}{2} - u\right) du \\
 &= \int_0^{\frac{\pi}{2}} \ln \cos u \, du \\
 &= \int_0^{\frac{\pi}{2}} \ln \cos x \, dx
 \end{aligned}$$

relabelling u as x .

(ii)

2

$$\begin{aligned}
 I &= \int_0^{\frac{\pi}{2}} \ln \sin x \, dx \\
 &= \int_{\pi}^{\frac{\pi}{2}} \ln \sin(\pi - u) \times -du \\
 &= \int_{\frac{\pi}{2}}^{\pi} \ln \sin(\pi - u) \, du \\
 &= \int_{\frac{\pi}{2}}^{\pi} \ln \sin u \, du \\
 &= \int_{\frac{\pi}{2}}^{\pi} \ln \sin x \, dx
 \end{aligned}$$

Hence

$$\begin{aligned}
 \int_0^{\pi} \ln \sin x \, dx &= \int_0^{\frac{\pi}{2}} \ln \sin x \, dx + \int_{\frac{\pi}{2}}^{\pi} \ln \sin x \, dx \\
 &= 2I
 \end{aligned}$$

The result now follows on dividing by 2.

(iii)

1

$$\begin{aligned}
 2I &= \int_0^{\frac{\pi}{2}} \ln \sin x \, dx + \int_0^{\frac{\pi}{2}} \ln \cos x \, dx \\
 &= \int_0^{\frac{\pi}{2}} \ln (\sin x \cos x) \, dx \\
 &= \int_0^{\frac{\pi}{2}} \ln \left(\frac{1}{2} \sin 2x \right) dx
 \end{aligned}$$

(iv) Use the substitution $u = 2x$ in part (iii). Then

2

$$\begin{aligned}
 2I &= \int_0^{\frac{\pi}{2}} \ln\left(\frac{1}{2} \sin 2x\right) dx \\
 &= \int_0^{\pi} \ln\left(\frac{1}{2} \sin u\right) \frac{1}{2} du \\
 &= \frac{1}{2} \times \int_0^{\pi} (\ln \frac{1}{2} + \ln \sin u) du \\
 &= \frac{1}{2} \times \left(\int_0^{\pi} \ln \frac{1}{2} du + \int_0^{\pi} \ln \sin u \right) du \\
 &= \frac{\pi}{2} \ln \frac{1}{2} + I \\
 &= -\frac{\pi}{2} \ln 2 + I
 \end{aligned}$$

Hence $I = -\frac{\pi}{2} \ln 2$.

(v) By implicit differentiation,

2

$$\begin{aligned}
 \frac{2x}{a^2} + \frac{2yy'}{b^2} &= 0 \\
 y' &= \frac{-xb^2}{ya^2} \\
 y' &= \frac{-x_1b^2}{y_1a^2} \quad \text{at } (x_1, y_1)
 \end{aligned}$$

The tangent at P has equation

$$\begin{aligned}
 (y - y_1) &= \frac{-x_1b^2}{y_1a^2} (x - x_1) \\
 \frac{yy_1 - y_1^2}{b^2} &= \frac{-xx_1 + x_1^2}{a^2} \\
 \frac{xx_1}{a^2} + \frac{yy_1}{b^2} &= \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} \\
 \frac{xx_1}{a^2} + \frac{yy_1}{b^2} &= 1
 \end{aligned}$$

since (x_1, y_1) is on $\mathcal{E}$.

(vi) We are given that gradient $AO = \text{gradient } AO'$ and required to show that gradient $PQ = \text{gradient } P'Q'$.

2

By rearranging the chord of contact formula, the gradient of PQ is $\frac{-x_0b^2}{y_0a^2}$.

Similarly the gradient of $P'Q'$ is $\frac{-x'_0b^2}{y'_0a^2}$.

Thus it is enough to show that $\frac{x_0}{y_0} = \frac{x'_0}{y'_0}$.

But gradient $AO = \frac{x_0 - 0}{y_0 - 0} = \frac{x_0}{y_0}$ and gradient $A'O = \frac{x'_0 - 0}{y'_0 - 0} = \frac{x'_0}{y'_0}$.

Since by assumption gradient $AO = \text{gradient } AO'$, we are done.

- (vii) The equations of the chords of contact PQ and $P'Q'$ from A and A' respectively are 2

$$\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1 \quad \text{and} \quad \frac{xx'_0}{a^2} + \frac{yy'_0}{b^2} = 1$$

Now since $R(x_3, y_3)$ is on the intersection of PQ and $P'Q'$, then it lies on both lines, that is its coordinates satisfy both equation. Hence

$$\frac{x_3x_0}{a^2} + \frac{y_3y_0}{b^2} = 1 \quad \text{and} \quad \frac{x_3x'_0}{a^2} + \frac{y_3y'_0}{b^2} = 1$$

But the chord of contact from $R(x_3, y_3)$ has equation $\frac{x_3x}{a^2} + \frac{y_3y}{b^2} = 1$.

By the previous two equations (x_0, y_0) and (x'_0, y'_0) lie on this line. Hence the line AA' is the line through the chord of contact from R .

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